To-Do List

1. Theoretical & Methodological Strengthening

- Formally state the assumptions under which the first-order perturbation approximation holds, especially for dynamically evolving graphs.
- Add a dedicated section comparing AEI to existing related concepts (effective resistance, edge-based spectral centrality, gradient-based smoothness measures) to clearly justify novelty.
- Add discussion of AEI's invariance properties and stability under noise.

2. Methodological Justification (Ablation Studies)

- Justify the choice of Pearson correlation for building the neuron co-activation graph.
- Run ablations comparing alternative graph construction methods (mutual information, cosine similarity) to show the results aren't sensitive to this design choice.

3. Baseline Comparisons

- Implement and compare against magnitude-based pruning.
- Implement and compare against L1/L2 norm pruning.
- Implement and compare against SNIP and GraSP (as suggested by Reviewer 3).

4. Computational Overhead Analysis

- Measure and report the overhead costs of graph construction (correlation matrix computation) and eigendecomposition (Fiedler vector) for all datasets.
- Report FLOPs and inference latency before and after pruning to justify the method's practical value.

5. Extended Experiments on Larger Architectures

- Apply the pruning method to at least one CNN architecture on a more challenging benchmark (e.g., CIFAR-10).
- Attempt at least a preliminary extension to a Transformer or report results on a larger-scale dataset to demonstrate scalability.

6. Statistical Rigor in Results

- Add confidence intervals to all accuracy comparisons.
- Add hypothesis testing (e.g., t-tests) to validate that spectral pruning is statistically significantly better than baselines.
- Include variance analysis across pruning runs.

7. Jazz Musician Network — Strengthen the Narrative

- Reframe or restructure this experiment so it more explicitly connects to and motivates the neural pruning results, **or** reduce its scope and move it to a supplementary/illustrative role.

8. Writing & Presentation Revisions

- Shorten and tighten the Introduction — remove or condense general network science content not directly relevant to the pruning contribution.
- Fix verbose, repetitive, and grammatically inconsistent passages throughout (especially Section 1 and around Line 110).
- Simplify Figure 1 — it currently tries to capture too much at once.
- Simplify other overly dense diagrams where possible.